Study: Mach-Zehnder Interferometer Circuit, SOI (DRAFT)

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I. Introduction

PHOTONIC integrated circuits (PICs) are an essential building block for the future of computing. The Mach-Zehnder interferometer (MZI) serves as one of the fundamental components on PICs, allowing for a single source of light to be self-interfered or for the interference of two distinct beams of light, utilising the relative phase shift between them. In general, the optical path length between the two arms of the interferometer are different and sometimes variable. The most simple way to accomplish this is to vary the physical length between the paths while keeping the refractive index constant. For variability, either electro-optic or thermo-optic modulators can be used.

Using the aforementioned modulators, MZIs can be constructed using 2x2 splitters and configured in such a way to create an optical NxN switch, allowing the efficient transfer of data around a PIC. In this report, for simplicity and to suit the scale of the assignment, we consider a 3x3 switch with static path length differences rather than variable optic modulators.

II. Theory

To construct an MZI, waveguides are used to direct the light towards two paths of different optical lengths. The paths are then recombined and the light interaction which occurs depends on the relative phases. For the basic theory of an MZI, consider another fundamental component, the Y-Branch.

The Y-Branch comes in two designs which are similar but distinct; the splitter and the combiner. The splitter has one path divided into two. If the intensity of the input is I_i, each output takes half of the intensity, $I_1 = I_2 = I_i/2$. For the electric field, $I_i \propto E^2$. Therefore at the output:

$$E_1 = E_i/\sqrt{2} \tag{1}$$

Similar theory applies to the combiner, where the output electric field is given by:

$$E_o = E_1/\sqrt{2} \tag{2}$$

in the case of a single input, or

$$E_o = \frac{E_1 + E_2}{\sqrt{2}} \tag{3}$$

in the case where both inputs are used.

Now consider the case where first light is split using a y-branch splitter, propagates along waveguides with a difference in length ΔL, then combined using a y-branch coupler. Electric field propagation, in general, is described by the plane wave equation:

$$E = E_i e^{i(\omega t - \beta z)} \tag{4}$$

where $\beta = \frac{2\pi n_{eff}}{\lambda}$ is the propagation constant and z is the distance travelled.

In the general case, we consider the two paths at the output of the splitter may have different propagation characteristics (β) and different loss coefficients (α).

For path lengths L_1 and L_2, we write the electric fields at the end of the paths, just before the combiner, as:

$$E_{o1} = E_1 e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \text{ and} \tag{5}$$

$$E_{o2} = E_2 e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \tag{6}$$

The splitter output, therefore, is:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}(e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}) \tag{7}$$

The intensity is then given by:

$$I_o = \frac{I_i}{4}|e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}| \tag{8}$$

If we consider the simpler case of equal propagation constants and no loss:

$$I_o = \frac{I_i}{2}[1 + cos(\beta \Delta L)] \tag{9}$$

For a single output device, the remainder of the light $(1 - cos(\beta \Delta L))$ is radiated away. However, if we consider a 2x2 coupler instead, that light is simply directed towards the other output port.

From this we can calculate the difference in path length required to split the light in the ratio I_o out of the bottom port of the interferometer and $I_i - I_o$ out of the top port. A 50/50 split ($I_0 = \frac{I_i}{2}$) at 1550nm in silicon, for example, will yield a ΔL as follows:

$$\Delta L = \frac{arccos(\frac{2I_0}{I_i} - 1)}{\beta} \tag{10}$$

$$\Delta L = \frac{\pi}{2\beta} \tag{11}$$

$$\Delta L = 155nm \tag{12}$$

III. Waveguide Analysis

The propagation constant mentioned previously can vary depending on the physical structure of the waveguide. Lumerical Mode software can be used to calculate various properties of waveguides including propagation of different components of electric and magnetic fields, propagation of different modes (TE and TM) and values for effective indices.

For SOI (Silicon on Insulator) waveguides, a common standard for 1550nm light is a 500nm wide by 220nm tall waveguide. The electric field intensity for this waveguide is shown in figures 1 and 2 for the first TE and TM modes, respectively.

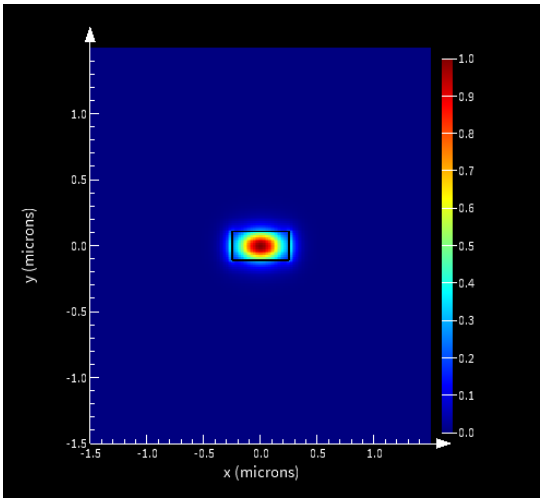


Fig. 1. Electric field intensity of the first TE mode of a 500 x 220 nm silicon waveguide for light at 1550 nm.

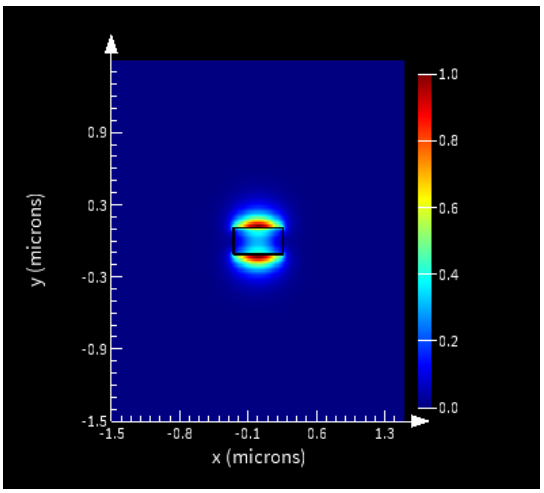


Fig. 2. Electric field intensity of the first TM mode of a 500 x 220 nm silicon waveguide for light at 1550 nm.

If the waveguide is made 100nm narrower or wider, not much of a change in the electric field is seen, however, there is a slight change in effective index (see table ref). More interestingly, if the waveguide is made slightly taller, it can be seen that the first TM mode can now be accommodated inside the waveguide and rather than the bulk of the electric field being outside the waveguide (fig. 2) it is contained and can propagate (fig. 3). This gives insight into the reasoning for the standard size; to isolate only the first TE mode (or only TE modes).

The effective index can be calculated from Lumerical Mode over a range of wavelengths, giving insight into how the waveguide will respond when there are frequency fluctuations in the light source. The index data can also be used to create a "compact model" of the waveguide, which is useful for transferring a strong approximation of the waveguide behaviour more easily. The compact model in general is a Taylor expansion of the effective index around a center wavelength, with the following format:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{13}$$

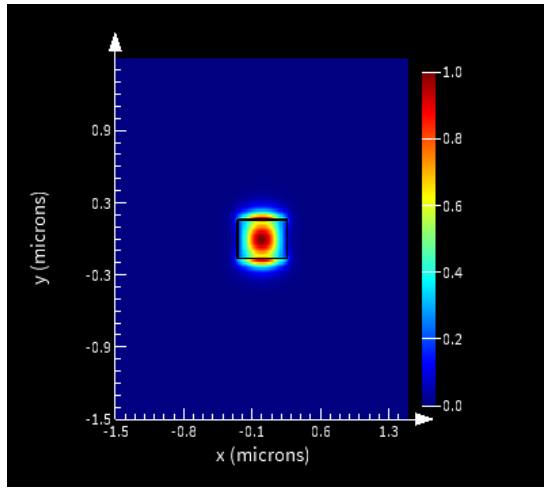


Fig. 3. Electric field intensity of the first TM mode of a 500 x 320 nm silicon waveguide for light at 1550 nm.

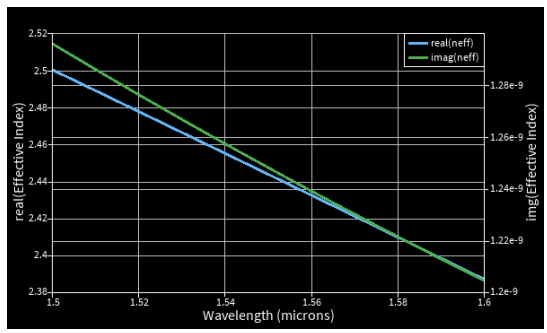


Fig. 4. The real and imaginary components of the effective index for the first TE mode of the 500 x 220 nm waveguide over a range of wavelengths. Note the y-axis scale for the imaginary component is on the right hand side.

Using a matlab script, the data from lumerical can be fitted to the compact model equation, giving the compact model equation for the 500 x 220 nm waveguide:

$$n_{eff}(\lambda) = 2.44391 + 1.13177(\lambda - 1.55) - 0.0426968(\lambda - 1.55)^2 \tag{14}$$

where the wavelengths are given in microns.

Lumerical Mode calculates many other important properties of the waveguide under simulation alongside the effective index, such as the group index (5), the loss per centimetre and the dispersion. While for very short and simple circuits, these values may not have large effects, for more complex projects these waveguide attributes must be carefully considered.

IV. Mach-Zehnder Interferometer

In this section, we consider MZIs with 2 inputs and 2 outputs, therefore composed of two 2x2 couplers with waveguides of differing lengths in between them. The couplers add some phase along the coupling length during crossover. This added phase can affect the output of the MZI, so to mitigate this, the second coupler is flipped horizontally and vertically. This has the effect of adding phase to the bottom path in the first coupler and adding the same amount of phase to the top path in the second coupler, negating the additions. We can then base our MZI output calculations on equation 9.

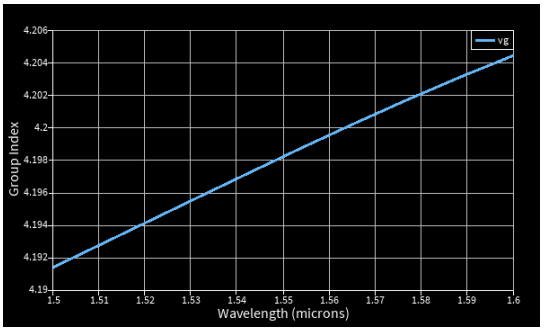


Fig. 5. The group index of the 500 x 220 nm waveguide.

An example of the gain for both of the outputs for an MZI with $\Delta L = 100nm$ is shown below in figure 6. As the gain of one port increases, the other decreases, leading to opposite maximums and minimums. The peaks are regularly spaced; the distance between them is called the free spectral range (FSR). The FSR of an MZI can be determined by the following equation:

$$FSR = \frac{\lambda^2}{n_g \Delta L} \tag{15}$$

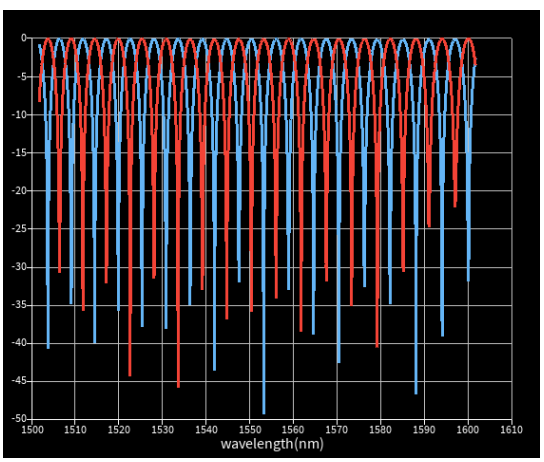


Fig. 6. The gain at both of the output ports of and imbalanced MZI with $\Delta L = 100nm$ over a broad spectrum.

For the purposes of a 3x3 splitter, we would ideally like a flat response to accommodate a broader range of wavelengths. To achieve this, we minimize ΔL, making the FSR as large as possible. Examining the transfer function equation (equation 9), the maximum is reached when the cosine term is equal to 1, and the minimum when that term is -1. This tells us the argument of the cosine term, which is the change in phase, functionally ranges from 0 to π (i.e. this range covers all possible power splits between ports). This in turn gives us the fundamental range of values for $\Delta L : [0, \sim 320]$. To minimize the FSR, ideally only very small differences in wavelength will be used, however, there are generally manufacturing constraints limiting this. In the most ideal case, we would additionally add phase changes to the end of the circuit to line up the phases of all three outputs.

V. Interconnect and KLayout Implementation

The 1×3 optical switch was modeled in *Lumerical INTERCONNECT* to validate the circuit logic and subsequently translated to a physical mask in *KLayout*.

A. Circuit Simulation and Spectral Analysis

To ensure phase symmetry and robust routing, a symmetric MZI architecture was adopted within the INTERCONNECT environment. As shown in Fig. 7, the circuit utilizes cascaded stages of 2×2 bidirectional couplers (BDCs). To mitigate the intrinsic phase shift added during crossover events—which was measured in the ONA at approximately 1.39 rad for a standalone BDC—the second coupler in each interferometer stage is flipped both horizontally and vertically. This configuration ensures that the intrinsic coupling phase is neutralized by adding phase to the bottom path in the first coupler and an equal amount to the top path in the second, allowing the output intensity to be governed primarily by the intentional physical path length difference ΔL.

The simulation utilized an Optical Network Analyzer (ONA) to verify the phase relationship across a region of the C-band. By maintaining the total phase shift within the $[0, \pi]$ range, the device covers all possible power splits between ports.

The spectral response was analyzed to distinguish between device performance and fiber-to-chip coupling effects. The raw gain response exhibits a parabolic profile characteristic of the grating couplers. By de-embedding the grating coupler response, the intrinsic circuit transmission is revealed to be remarkably flat across the C-band [1540-1560 nm] (Fig. 8). This broadband performance is a direct result of minimizing ΔL, which maximizes the Free Spectral Range (FSR) and ensures functionality over a wide operating window.

B. Physical Layout

The physical translation in *KLayout*, illustrated in Fig. 9, adheres to the 500×220 nm SOI waveguide standard. The static phase shifts are implemented through localized waveguide bends that create precise length differences. For this design, the actual range of ΔL values needed to achieve the required phase shifts at a center wavelength of 1550 nm was $[0, \sim 264]$ nm.

To ensure a flat spectral response and accommodate a broader range of wavelengths, ΔL was minimized to maximize the Free Spectral Range (FSR). The layout accounts for specific bend radii to minimize radiation loss while ensuring that the first TM mode remains poorly confined compared to the primary TE mode.

VI. 1x3 Switch Transfer Function Derivation

To accurately model the behavior of the cascaded 1×3 architecture, we must track the complex electric field propagation through the circuit to preserve the relative phase information between stages. We define the wavelength-dependent phase delay introduced by the physical path length difference of the k-th MZI as $\phi_k = \frac{\beta \Delta L_k}{2}$, where β is the propagation constant of the waveguide.

Assuming ideal 50/50 coupling, the unitary transfer matrix M_k for a single passive MZI is given by:

$$M_k = e^{-i\phi_k} \begin{pmatrix} -i\sin(\phi_k) & -i\cos(\phi_k) \\ -i\cos(\phi_k) & i\sin(\phi_k) \end{pmatrix} \tag{16}$$

For the specific Clements-inspired routing of this 3x3 switch, configured to operate as a 1x3 divider, the entirety of the input field E_{in} is directed into the top port of MZI 1. The bottom port of MZI 1 receives no input. The outputs of MZI 1 subsequently feed the next layer of the mesh: the top output routes to the top port of MZI 3, while the bottom output routes to the top port of MZI 2. Finally, the top output of MZI 2 feeds the bottom port of MZI 3, and the bottom output of MZI 2 serves as Channel 3.

We can express the total system transfer function as the cascaded product of the individual stage matrices. To accommodate the 3-port

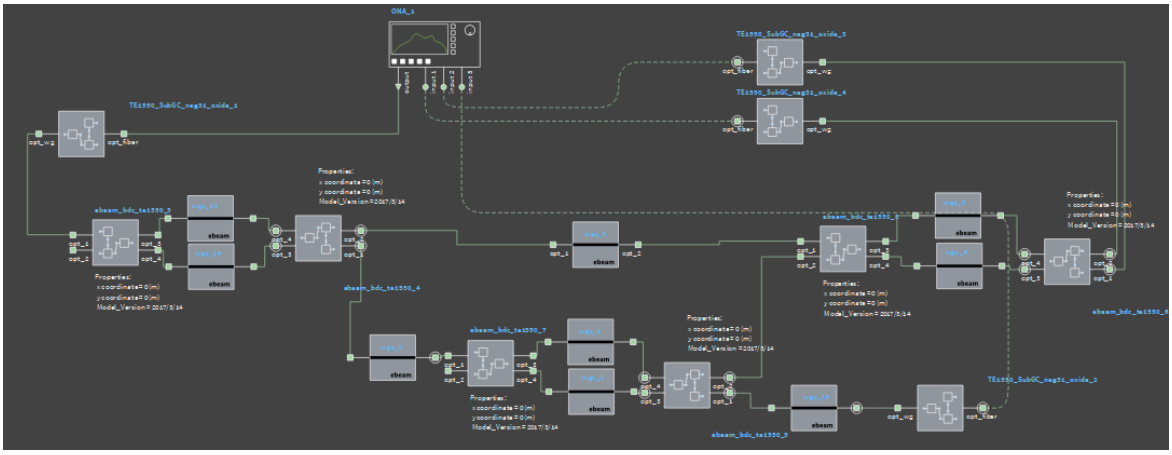
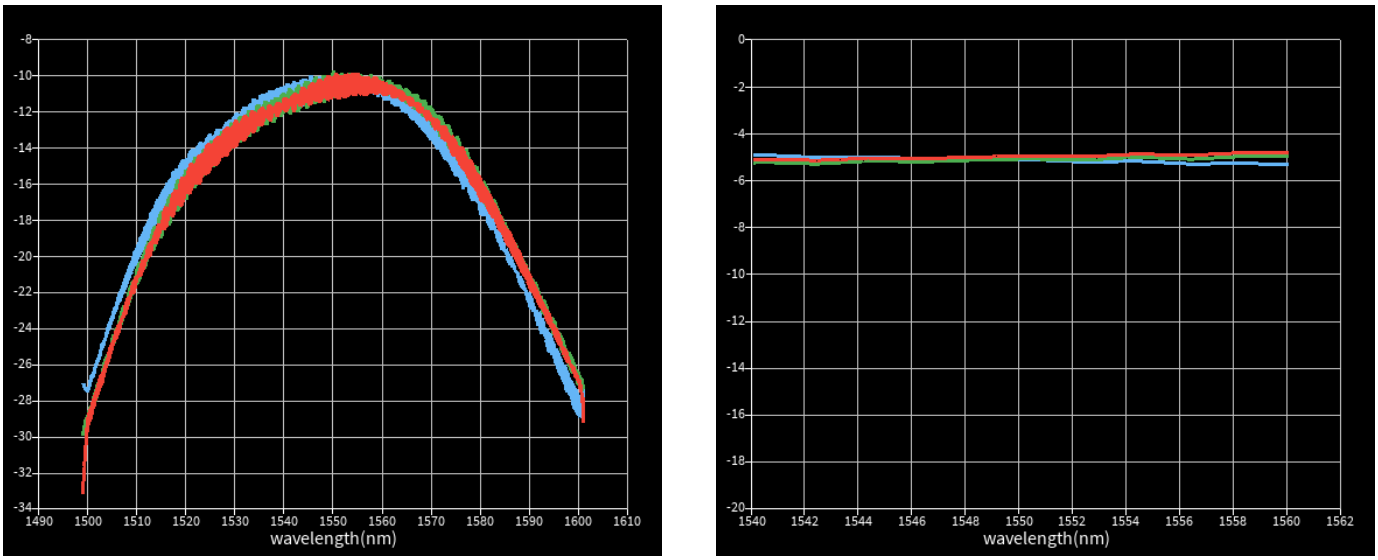


Fig. 7. Full INTERCONNECT circuit schematic of the 1-to-3 switch. The cascaded symmetric MZI stages allow for wavelength-independent routing logic by neutralizing coupler-induced phase offsets.



(a) Gain with Grating Couplers

(b) Intrinsic Circuit Gain (De-embedded)

Fig. 8. Spectral response comparison. The raw data (left) shows the Gaussian-like envelope of the grating couplers, while the de-embedded data (right) demonstrates the broadband, flat response of the MZI circuit achieved by maximizing the FSR.

routing, the 2×2 MZI matrices are expanded into a single topological equation:

$$\begin{pmatrix} E_{out,1} \\ E_{out,2} \\ E_{out,3} \end{pmatrix} = \begin{pmatrix} M_{11}^{(3)} & M_{12}^{(3)} & 0 \\ M_{21}^{(3)} & M_{22}^{(3)} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & M_{11}^{(2)} \\ 0 & M_{21}^{(2)} \end{pmatrix} \begin{pmatrix} M_{11}^{(1)} \\ M_{21}^{(1)} \end{pmatrix} E_{in} \tag{17}$$

By evaluating this matrix multiplication and substituting the trigonometric terms from the passive MZI matrix, we obtain the explicit electric field equations for the three output channels:

$$E_{out,1} = \Big[-\sin(\phi_1)\sin(\phi_3)e^{-i(\phi_1+\phi_3)} + i\cos(\phi_1)\sin(\phi_2)\cos(\phi_3)e^{-i(\phi_1+\phi_2+\phi_3)} \Big] E_{in} \tag{18}$$

$$E_{out,2} = \Big[-\sin(\phi_1)\cos(\phi_3)e^{-i(\phi_1+\phi_3)} - i\cos(\phi_1)\sin(\phi_2)\sin(\phi_3)e^{-i(\phi_1+\phi_2+\phi_3)} \Big] E_{in} \tag{19}$$

$$E_{out,3} = -\cos(\phi_1)\cos(\phi_2)e^{-i(\phi_1+\phi_2)} E_{in} \tag{20}$$

It is critical to note the implications of this transfer function on experimental parameter extraction. As discussed in previous sections, the primary design objective for this 1x3 switch was to achieve a highly flat spectral response across the target operational bandwidth ($\sim$ 1540 - 1560 nm). To accomplish this, ΔL was intentionally minimized, which consequently maximizes the Free Spectral Range (FSR). Because the resulting FSR is significantly larger than the measurement window, the transmission spectra present as single, broad plateaus rather than periodic fringes.

As a result, standard data analysis techniques reliant on the FSR—such as peak-finding algorithms traditionally used to extract the group index (n_g) or perform automated curve fitting—are not applicable to this dataset. The mathematical degeneracy introduced by the exceptionally flat response prevents the robust extraction of dispersion parameters solely from the transmission amplitude, necessitating the use of fixed theoretical dispersion constraints during regression analysis.

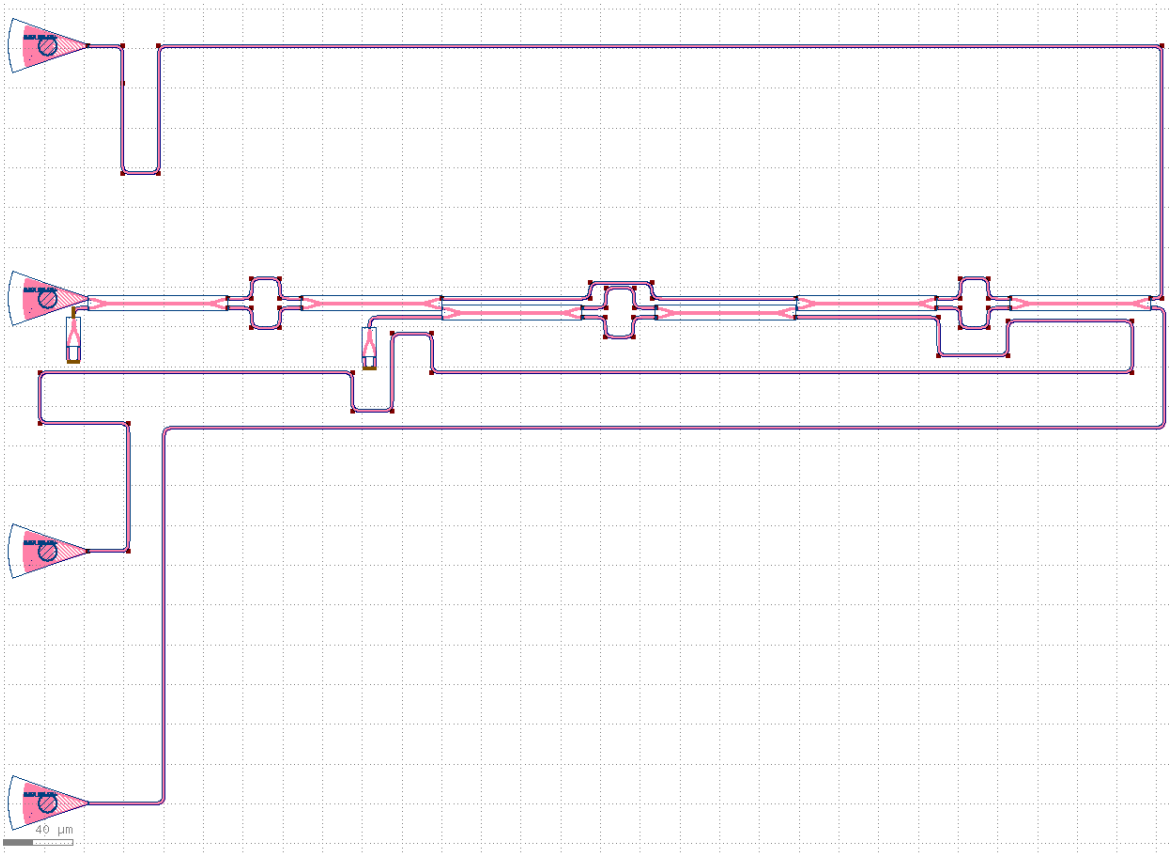


Fig. 9. Complete KLayout GDSII mask design. The layout utilizes 500×220 nm waveguides to isolate the TE mode while precisely controlling arm lengths for static switching.

VII. Corner Analysis and Fabrication Variability

Photonic integrated circuits are highly sensitive to nanometer-scale variations inherent in the fabrication process. Tolerances in lithography and etching processes inevitably lead to deviations in the final width and height of the silicon waveguides across a wafer. To account for this variability and ensure robust circuit operation, a corner analysis is typically performed by simulating the optical properties of the waveguide at the extreme boundaries of the expected manufacturing tolerances.

Table I presents the fundamental waveguide parameters extracted for various dimensional extremes. These values were calculated using the Lumerical MODE solver to simulate physical variations around the nominal 500×220 nm waveguide cross-section.

TABLE I
Waveguide Parameters for Fabrication Corner Analysis
($\lambda_0 = 1550nm$)

Waveguide Dimensions (nm)	$\mathbf{n_{eff}}$	$\mathbf{n_g}$	D (ps/nm/km)
500×220.0	2.44435	4.1959895	432.0
510×215.3	2.44229	4.1754225	329.0
510×223.1	2.47100	4.1824480	468.0
470×223.1	2.40380	4.2547945	521.0
470×215.3	2.37473	4.2463395	330.0

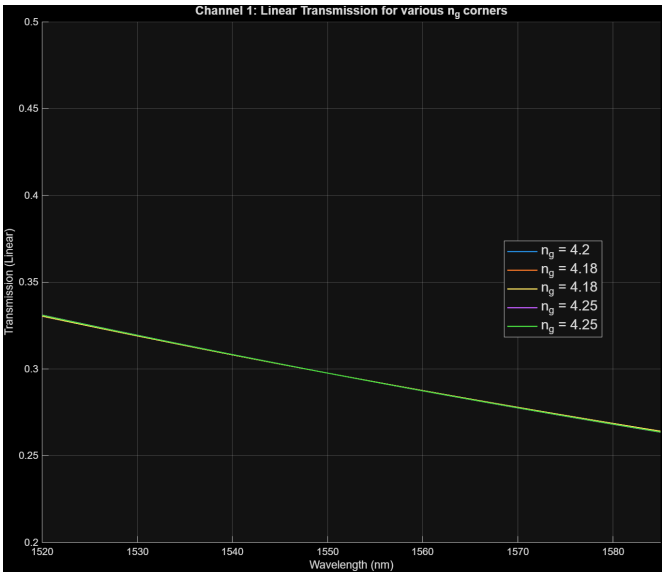
To evaluate the functional impact of these physical variations on the final device, the extracted values for the effective index (n_{eff}) and group index (n_g) are substituted directly into the previously derived 1×3 switch transfer function. Because the phase delay terms (ϕ_k) governing the interferometric behavior are directly proportional to n_{eff}, passing these corner values through the transfer equations allows us to systematically view the impact of the group index on the circuit. This reveals the worst-case spectral shifts and power splitting imbalances the switch might experience post-fabrication.

It should be noted that while this corner analysis effectively models uniform cross-sectional deviations, it is not an exhaustive representation of all physical imperfections. In physical implementations utilizing long routing waveguides, it is highly likely that there will be additional, compounding phase deviations caused by systemic wafer-level thickness gradients and localized sidewall roughness. These highly localized effects can have a great impact on expected performance of a circuit and typically require more advanced modeling to quantify. In the model analysis given here, for example, the circuit under test has seemingly extraneous lengths and bends in the output waveguides, which are an attempt to synchronize the relative phases at the outputs. These extra lengths would compound effects due to variations in the group index while incurring further changes due to local effects across the greater span of the waveguides.

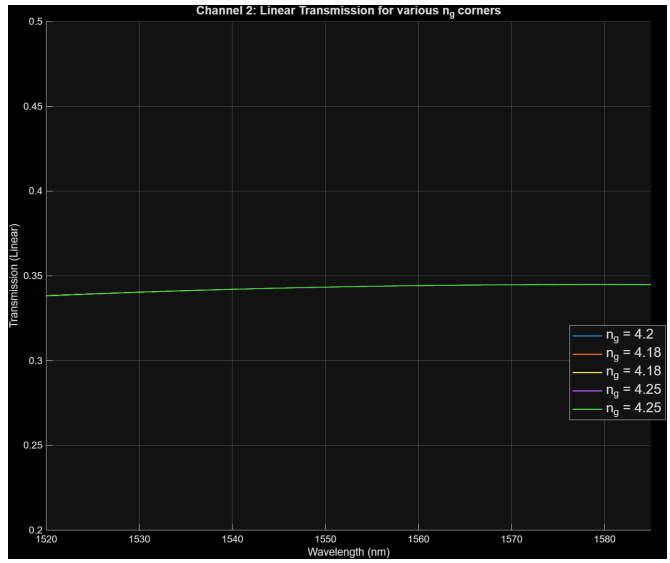
In figure 10, we see a small deviation in transmitted power over the wavelength range, with Output 1 losing power and Output 3 gaining power as the wavelength increases. Output 2 is relatively stable across the range and the split between all three outputs at the target wavelength of 1550nm is very close to even. There are no noticeable differences between simulations with different group indices, as there may be for an MZI with a short FSR.

VIII. Measured Output and Performance Discrepancies

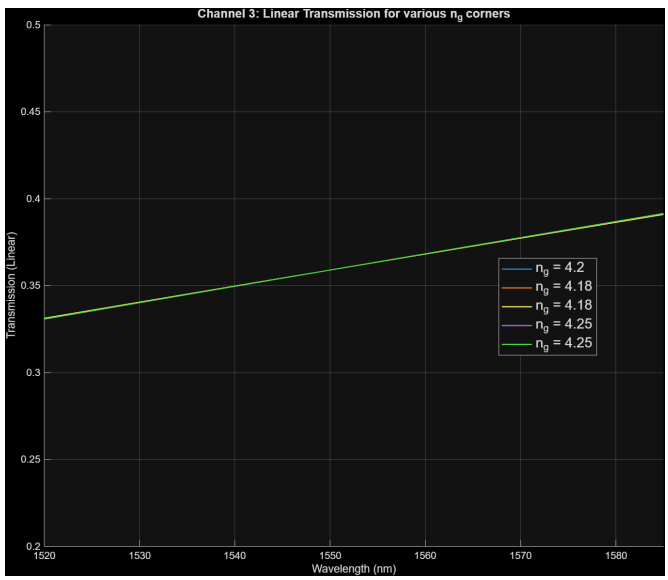
As shown in the system transmission response (Fig. 11), the measured spectral output of the fabricated 1×3 switch is significantly weaker than the theoretical simulations. The empirical data, which



(a) Output 1 Corner Analysis



(b) Output 2 Corner Analysis



(c) Output 3 Corner Analysis

Fig. 10. Corner analysis results demonstrating the impact of fabrication variability across all three output channels.

was acquired from the IRIS testing stage, reveals severe attenuation with transmission levels dropping to between -20 dB and -45 dB alongside pronounced high-frequency oscillations. We also observe the characteristic bowed shape of the grating couplers.

This substantial discrepancy in insertion loss can be attributed to a combination of fabrication realities and experimental limitations. Primary contributors include intrinsic manufacturing defects and inherent signal losses within the optical testing fibers. Furthermore, fiber-to-chip coupling at the grating couplers is highly sensitive; minor misalignments or any motion during the measurement process can introduce severe insertion penalties. Finally, as noted in previous sections, the physical routing of the layout necessitated additional waveguide lengths that were not explicitly accounted for in the baseline simulations. This extra propagation distance inherently increases optical loss and drastically compounds the circuit's susceptibility to wafer-level dimensional variations, contributing to the degraded performance observed in the measured data.

The most notable measurement result is the "Measured 1" curve from the top output of the system, which sits about 10dB below the other outputs. While it's possible that this is due to an alignment error on the IRIS stage, it is probably due to a design error in the circuit; where the top output of MZI 1 bypasses MZI 2 there are "S" curves in the U-bend. It's possible the corner radii are too tight and light is radiated out here, affecting the output. However, if this is the case, there would be an expectation of an error in the middle output, "Measured 2", as they are mixed in MZI 3. This doesn't appear to be the case. Note also that the Scylla measuring stage didn't measure any output from the top output.

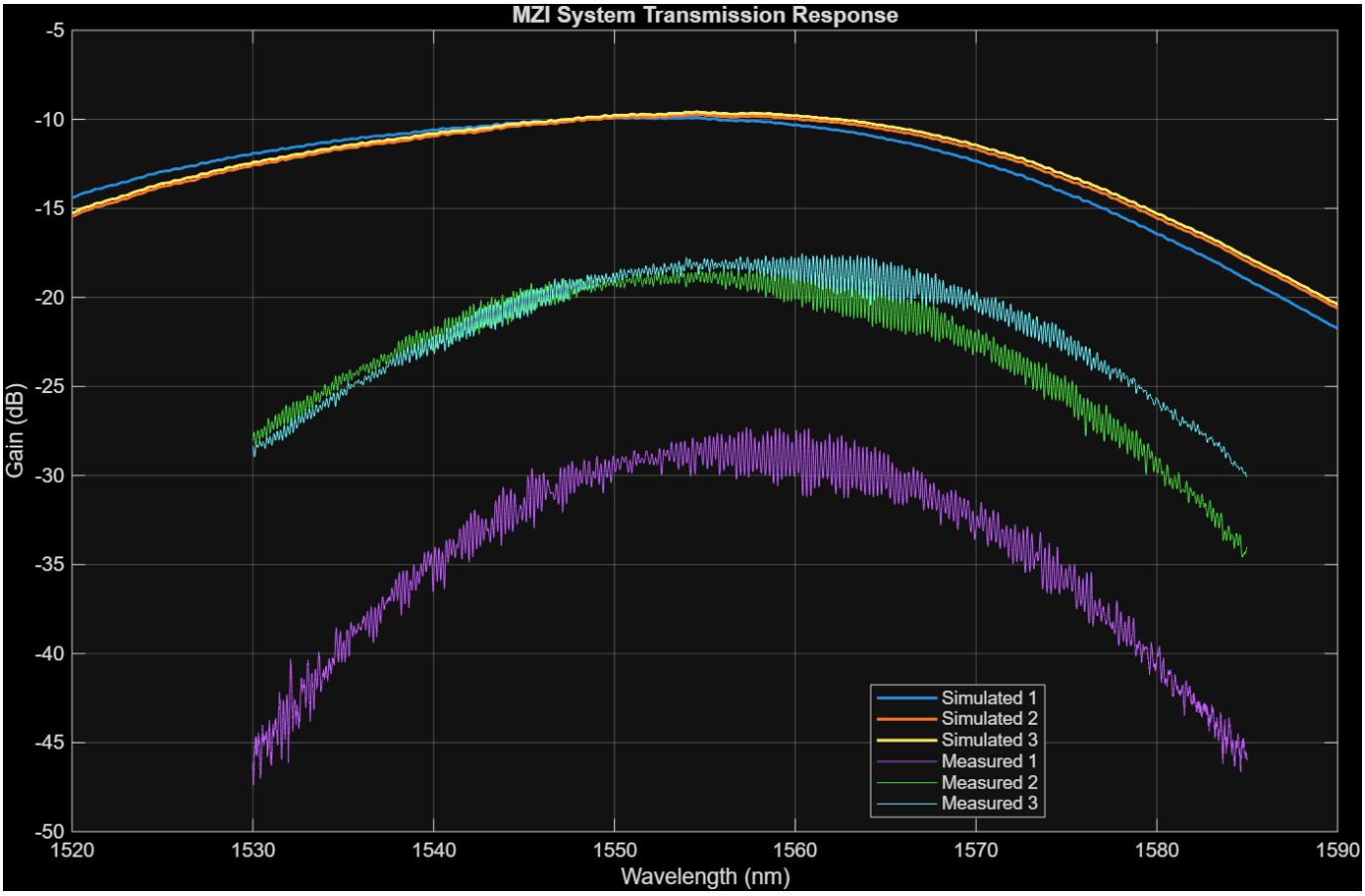


Fig. 11. System transmission response comparing simulated outputs against the empirical data obtained during the IRIS testing stage. The measured response exhibits significant insertion loss and high-frequency noise relative to the ideal simulation.

IX. Baseline Corrected Data

Raw transmission measurements from photonic integrated circuits inherently include the wavelength-dependent insertion loss of the input and output grating couplers. These couplers typically exhibit a strong Gaussian or parabolic transmission envelope that obscures the intrinsic optical response of the underlying circuit.

To isolate the performance of the 1×3 MZI mesh, a baseline correction procedure was applied to the empirical data. The dataset was first truncated to the primary region of interest (1530 nm to 1585 nm) to eliminate extreme edge noise. A 6th-order polynomial was then fitted to the raw spectrum to mathematically approximate the grating coupler insertion loss envelope. By subtracting this polynomial fit from the original measurements and normalizing the peak values to approximately 0 dB, the true (relative), flattened spectral response of the device was extracted.

The results of this baseline correction are shown in Fig. 12. While the macroscopic grating coupler envelope has been successfully removed—demonstrating the intended flat, broadband response of the minimized ΔL design—the corrected data further highlights the underlying issues with the physical circuit. Specifically, removing the envelope exposes the severe high-frequency noise and signal instability that stem from the aforementioned manufacturing variations, excess routing lengths, and coupling inefficiencies.

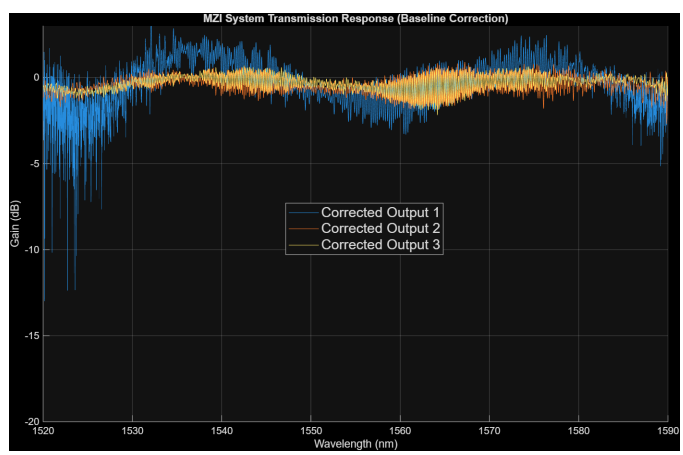


Fig. 12. Baseline-corrected system transmission response across all three channels. The 6th-order polynomial subtraction successfully flattens the grating coupler envelope, but clearly exposes the severe high-frequency noise and instability inherent to the fabricated device.